

Atharva Khedkar

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EDUCATION

Arizona State University

PhD - Computer Science

Tempe, Arizona

August 2023 - Present

- **GPA: 4.0.**
- **Courses:** Computer Architecture, Machine Learning Accelerator Design, Foundations of Algorithms.

G H Rasoni College of Engineering (GHRCE)

Bachelor of Technology - Artificial Intelligence

India

August 2019 - April 2023

- **CGPA: 9.33/10.**
- **Gold Medalist** (departmental) in freshman and junior years.

RESEARCH AND WORK EXPERIENCE

Arizona State University

Graduate Research Assistant

Tempe, Arizona

August 2023 - Present

Principal Investigator: **Dr. Aviral Shrivastava**

- Working on heterogeneous AI compiler to automatically partition ML workloads across devices.
- Developed a Digital Signal Processing Dialect ([DSP-MLIR](#)) using Multi-level Intermediate Representation (MLIR).

AMD

ML Intern

Hyderabad, India

February 2023 - July 2023

- Worked in Integration & Validation (IV) team and created ML networks to automate graphics driver fault detection.
- Successfully reduced test planning time by 80%, resulting in faster turnaround for driver releases while maintaining high-quality standards.

University of California San Diego

Research Intern

Remote

June 2022 - May 2023

Principal Investigator: **Dr. Tzyy-Ping Jung**

- Worked on EEG signals-based stress classification in classroom environment at the Swartz Center for Computational Neuroscience.
- Investigated and Developed machine learning algorithms to detect changes in stress levels based on the power spectral density of EEG Signals in classroom environment.

HackerRank

SDE Intern

Bangalore, India

July 2022 - December 2022

- Analyzed various ML Pipeline orchestration tools such as Pachyderm and MLflow and deployed AWS Lambda function to automate model deployments from MLflow Model registry into Sagemaker endpoints.
- Developed and Launched Application Tracker feature used by **1000+** developers on the [HackerResume](#) project.

RESEARCH PUBLICATIONS

[Google Scholar: bit.ly/akhgs]

- Kumar, Abhinav, **Atharva Khedkar**, Hwisoo So, Megan Kuo, Ameya Gurjar, Partha Biswas, and Aviral Shrivastava. "DSP-MLIR: A Domain-Specific Language and MLIR Dialect for Digital Signal Processing." In *Proceedings of the 26th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems*, pp. 146-157. 2025.
- **Khedkar, Atharva**, Atharva Peshkar, Ashlesha Nagdive, Mahendra Gaikwad, and Sudeep Baudha. "Exploiting spatiotemporal inconsistencies to detect deepfake videos in the wild." In *2022 10th International Conference on Emerging Trends in Engineering and Technology-Signal and Information Processing (ICETET-SIP-22)*, pp. 1-6. IEEE, 2022
- Patil, Darshan, Gopal Sakarkar, Pearl Khatri, **Atharva Khedkar**, Hong-Seng Gan, and Muhammad Hanif Ramlee. "Selective Attention UNet for Segmenting Liver Tumors." In *2023 International Conference on Cyber-Physical Social Intelligence (ICCSI)*, pp. 44-49. IEEE, 2023.

ACHIEVEMENTS & PROFESSIONAL MEMBERSHIPS

[LinkedIn: bit.ly/akhin]

- Winner of **Knowledge Catalyst Award** in the **Design Automation Conference (DAC) 2025**.
- Winner of the **Richard E. Merwin Student Scholarship** - Spring 2022 by IEEE Computer Society.
- **Microsoft Imagine Cup 2022** (Olympics of Technology) - **Asia Finalists** (Among Top 12 in Asia) and **National (India) Champion**.
- **Best Paper Award** at the 10th IEEE International Conference on Emerging Trends in Engineering & Technology Signal and Information Processing (IEEE ICETET SIP-22) "Exploiting spatiotemporal inconsistencies to detect deepfake videos in the wild".
- **Microsoft Imagine Cup 2021, World Finalists** (Among Top 39 in World) and **National Champion** (Top out of 10,353 project entries - 353 team and 10,000+ individual).
- **Winner** of [Accenture Applied Intelligence Hackathon](#) Organized by **Accenture India**.

- Runner up of Reimagine <Future> Hackathon Organized by **IBM**.
- Runner up of the AI for healthcare hackathon sponsored by the **Government of India**.

ACTIVITIES & LEADERSHIP

Entrepreneurship Cell - GHRCE

September 2021 - August 2022

Chairperson

- Promoted Startup culture while focusing on Female Entrepreneurs in Technology by organizing B-plan and business development workshops.
- Assisted in the incubation of 7+ startups and 42+ students at the Campus Business and Incubation Center (GHRTBIF).

IEEE CIS Student Branch Chapter - GHRCE

January 2021 - December 2021

Vice Chair

- Organized and Mentored 10+ events with 3000+ participants from all over the world.
- Under my vision, IEEE CIS SBC received the "2022 Outstanding Chapter Award" by IEEE CIS Awards Committee with the citation "For the innovative and wide range of diverse and inclusive activities which connect research, education and industry in the field of computational intelligence."

Microsoft Learn Student Ambassador

April 2020 - Jan 2025

Gold Level

- Trained more than 500 students by conducting a machine learning bootcamp as a student ambassador.
- Conducted 20+ sessions on open source, GitHub and mentored 500+ students in MLH Hackathons.

PERSONAL PROJECTS

[GitHub: bit.ly/akhgh]

DSP-MLIR

- **DSP-MLIR** is a framework consisting of a DSP-DSL, DSP-dialect, DSP optimizations and lowerings to standard MLIR dialects.
- **DSP-MLIR** outperforms standard compilation flows i.e. Clang, Hexagon-Clang and GCC on Qualcomm-hexagon NPU and CPU devices by **12%**.

Detect-d

- **Detect-d** is a sophisticated deepfake detection platform with **96.8 %** validation accuracy.
- Trained a CNN-LSTM model on the ImageNet dataset consisting of 14 million images in addition to the sophisticated datasets- Face Forensics++ and Deepfake Detection Challenge Dataset with about 7500 videos- containing both real and real-world deepfake videos.

SKILLS & CERTIFICATIONS

- **Key competencies:** Compiler Development, MLIR, Artificial Intelligence and Deep Learning.
- **Languages:** Advanced in C++, Python, and C.
- **Tools and Frameworks:** MLIR, LLVM, TensorFlow, and Microsoft Azure.
- **Languages:** Conversational Proficiency in English, Hindi, and Marathi.
- **Certifications:** DeepLearning.AI TensorFlow Specialization, Machine Learning (Coursera- Stanford University).

MEDIA COVERAGE

- These Nagpur based students are fighting Deepfakes with AI - **INDIAai (Govt. of India)**
[Link: <https://tinyurl.com/indiaai>]
 - Two teams from India proceed to the 2021 Microsoft Imagine Cup Finals- **Microsoft News (Microsoft Corp)**
[Link: <https://tinyurl.com/igcup>]
 - Young Innovators Bag Microsoft's Imagine Cup 2022 For Their Out Of The Box Ideas - **The Logical Indian**.
[Link: <https://tinyurl.com/migcup>]
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All Links